

Jason Wakim Richards

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Aerospace and GNC engineer with a First-Class MEng and hands-on experience developing, debugging, and deploying C/C++ and Python software for spacecraft ADCS, planetary-rover GNC simulation, and autonomous UAVs. Technical strengths include multisensor state estimation (EKF and visual-inertial odometry), attitude determination and control, rigid-body dynamics, motion planning, simulation and performance analysis, and GNC integration with sensors and vehicle systems. Led real-world LiDAR integration and testing for a TRL 6 rover demonstration, developed embedded CubeSat flight software, and now conduct simulation, laboratory, and flight testing of ROS 2/PX4 autonomy. Experienced with Linux, CMake, Git/GitLab, Docker, MATLAB/Simulink, technical documentation, and collaborative multi-vehicle autonomy.

Work Experience

Doctoral Researcher – Aerospace Autonomy & GNC Software

November 2025 – Present

SnT, University of Luxembourg | Kirchberg, Luxembourg

- Develop C++ and Python software for Semantic Relational SLAM and autonomous UAV systems, linking camera-IMU state estimation, perception, semantic mapping, navigation, planning, and control.
- Integrate, debug, and deploy onboard autonomy on physical UAVs using ROS 2, PX4, embedded Linux, Raspberry Pi, camera/IMU sensing, and OpenVINS visual-inertial odometry.
- Evaluate algorithm and system performance through simulation, laboratory, hardware, and flight testing; analyze estimation and autonomy data and trace failures across software, sensors, and vehicle interfaces.
- Build reproducible Docker-based development, deployment, and camera-IMU calibration workflows with Kalibr, supported by version-controlled software, design rationale, and technical documentation.
- Research fleet-level and multi-vehicle coordination using shared semantic maps, resource-efficient navigation, and mission-aware collaborative planning.

AOCS/GNC Engineer Intern

July 2023 – August 2024

Airbus Defence & Space | Stevenage, UK

- Led LiDAR test activities during Integrated BreadBoarding (iBB) 3 & 4 campaigns, supporting an end-to-end Sample Fetch Rover concept-of-operations demonstration at Technology Readiness Level 6.
- Designed a networked sensor test bench, integrated LiDAR with rover breadboards using ROS over a local wireless network, executed test procedures, and collected and analyzed campaign performance data.
- Developed and debugged C/C++ software on Linux for an offline GNC simulation, coupling a natural-scene generator to produce synthetic imagery and LiDAR scans for algorithm development and performance evaluation.
- Investigated terrain generation, LiDAR/radar sensor modeling, and Vector Field Histogram planning for lunar-rover applications, connecting simulation assumptions to navigation performance.
- Delivered software and hardware integration in a fast-paced Agile team using Git/GitLab, Jira, and Confluence; documented test status and resolved issues across GNC, simulation, sensors, and vehicle hardware.

Projects & Leadership

JamSail 3U CubeSat – AOCS/ADCS Lead

September 2024 – August 2025

NottSpace, University of Nottingham | Nottingham, UK

- Led ADCS design and implementation for a 3U CubeSat carrying a GNSS-interference sensing payload and new solar-sail design; translated the mission concept of operations into subsystem requirements, architecture, interfaces, and a verification plan.
- Developed embedded C flight software and a C/C++/CMake Linux simulation of rigid-body dynamics, attitude estimation, and control across mission modes.
- Implemented TRIAD and an EKF to fuse sun-sensor, magnetometer, and IMU measurements, together with sensor filtering and control-state logic for robust attitude determination.
- Implemented the International Geomagnetic Reference Field (IGRF) in C, converting an environmental model into a physics-based input for simulation, estimator development, and verification.
- Coordinated ADCS interfaces with the spacecraft team and applied ECSS-aligned software and verification practices across simulation and hardware testing.

Nottingham Space Society | Nottingham, UK

- Designed system architecture and subsystem interfaces within a student-led, 12-person Nottingham Space Society team developing a 3U CubeSat; led the requirements, Project Breakdown Tree, and Work Breakdown Structure.
- Developed software on a Raspberry Pi running Linux and integrated a microcontroller, IMU, and cameras before the concept progressed to the competition build phase.
- Contributed across embedded software, hardware and sensor integration, and system simulation while coordinating technical development within the student team.
- Achieved a top-four finish among 12 UK teams as first-time entrants to the UKSEDS Satellite Design Competition.
- Led cross-team meetings, communicated design changes, and presented the integrated concept to industry experts, balancing technical interfaces, build constraints, and competition deadlines.

UKSEDS Volunteer

Sept. 2024 – Sept. 2026

UK Students for the Exploration and Development of Space | Remote & UK field events

- Served as an Olympus Rover Trials (ORT) volunteer in the first year, helping organise a UK national rover competition with Airbus Defence & Space and RAL Space.
- Progressed in the second year to Competitions Management Co-Lead, overseeing four national competitions: ORT, SDC, ISAM, and NRC.
- Coordinated volunteers, schedules, technical milestones, participant support, and partner engagement with Airbus Defence & Space (SDC), the UK Space Agency (ISAM), Midlands Rocketry Club, and Skyrora (NRC).
- Produced clear participant documentation and supported venue and field-event logistics, helping multidisciplinary volunteer teams deliver competitions to fixed schedules.

Education & Training**MEng Aerospace Engineering – First-Class Honours**

September 2020 – August 2025

University of Nottingham | Nottingham, UK

- Graduated with First-Class Honours, with strong results in Dynamics & Flight Mechanics (99%), Avionics Systems (91%), and Introduction to Space (83%).
- Specialized in aerospace control systems, flight dynamics, avionics, spacecraft systems, modeling, optimization, and simulation.
- Applied C/C++, Python, MATLAB, Simulink, and Linux to state estimation, control, flight-mechanics analysis, and autonomous-system projects.

IEEE RAS Summer School on Multi-Robot Systems

July 2026

Czech Technical University in Prague | Prague, Czechia

- Completed the 2026 IEEE RAS Summer School on Multi-Robot Systems, covering cooperative aerial robotics and active perception; presented the SaCoGraph collaborative-UAV research concept.
- Applied multi-robot autonomy concepts to collaborative sensing, semantic mapping, and resource-aware mission planning.

Space Resources & Robotics Summer School

May 2026

AGH University of Kraków | Krakow, Poland

- Completed intensive training in space resources, planetary robotics, and robotic manipulation for exploration applications.
- Developed a broader systems perspective on robotic operations in challenging off-world environments.

Space Systems Engineering Training Course

May 2025

ESA Academy | Transinne, Belgium

- Completed intensive ESA Academy training across the space-system lifecycle, from mission objectives and concepts of operations through architecture, integration, verification, and operations.
- Applied requirements engineering and concurrent-design methods to translate user needs into traceable, testable requirements and manage interfaces between technical disciplines.
- Practised system options and trade studies, budgets and margins, verification and validation planning, and the relationship between engineering decisions, schedule, cost, and risk.
- Completed multidisciplinary group exercises and communicated design rationale and systems-engineering deliverables, strengthening collaborative problem-solving across complex spacecraft projects.